Steering LVLMs via Sparse Autoencoder for Hallucination Mitigation

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MOTIVATIONS & CONTRIBUTIONS

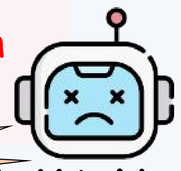


Please describe this image in detail.



The image captures a cozy living room with a brown sofa, two pillows, and a gray carpet under the sofa. A window, although not visible in the image.

Hallucinations



LLaVA-Next

Limitations of Prior Work:



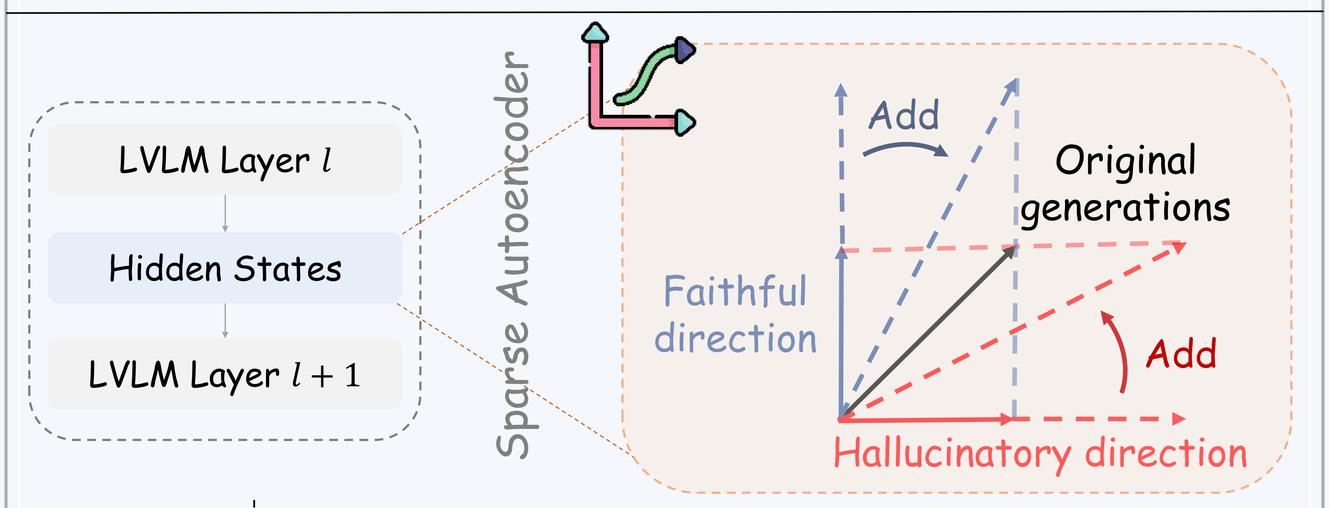
- **High Cost:** Methods relying on external knowledge, or decoding strategies are computationally expensive.
- **Ineffective:** Directly adjusting internal representations may be insufficient or might excessively intervene.

Our Solution: Using **SAE** to **identify and suppress** hallucination-related internal representations.



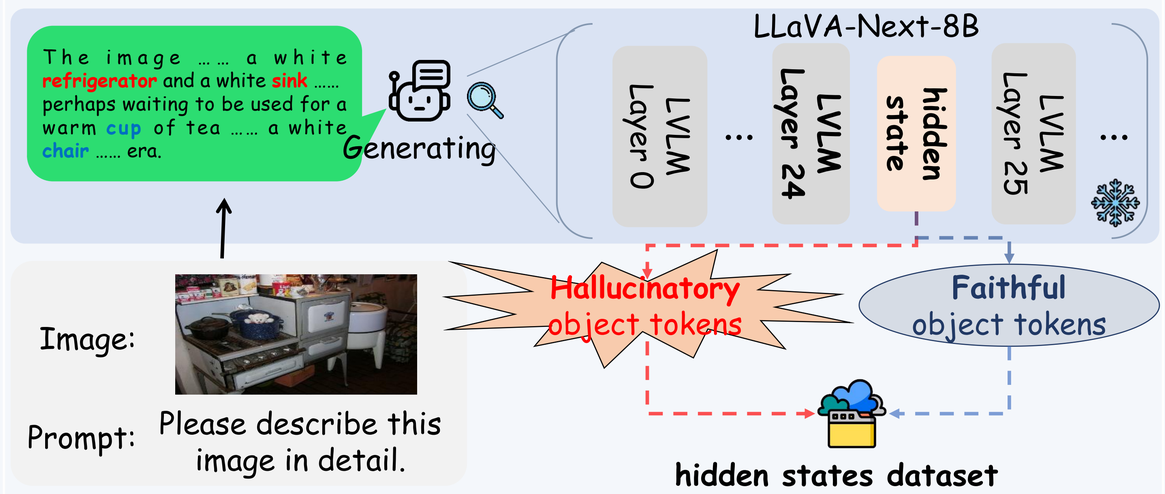
Contributions: ★ We first to use SAE provided by llm-labs to **identify semantic directions** for hallucination and faithfulness in LVLMs. ★ We proposed SSL, **a plug-and-play**, and generalizable method for hallucination mitigation. ★ SSL **outperforms** existing decoding methods across multiple models and benchmarks with **negligible overhead**.

INSIGHTS -- SEMANTIC DIRECTIONS

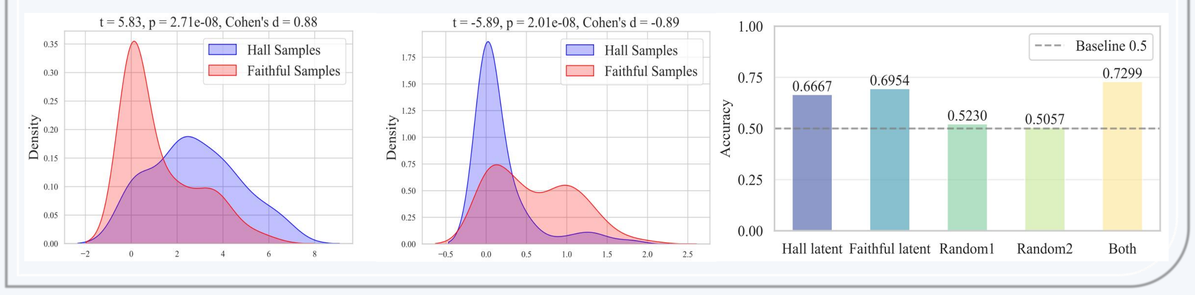


Insights: We leveraged SAE to extract semantic directions related to **hallucination** and **faithfulness**. We demonstrated that **steering** along the hallucinatory direction increases hallucination, while along the faithful direction mitigates it.

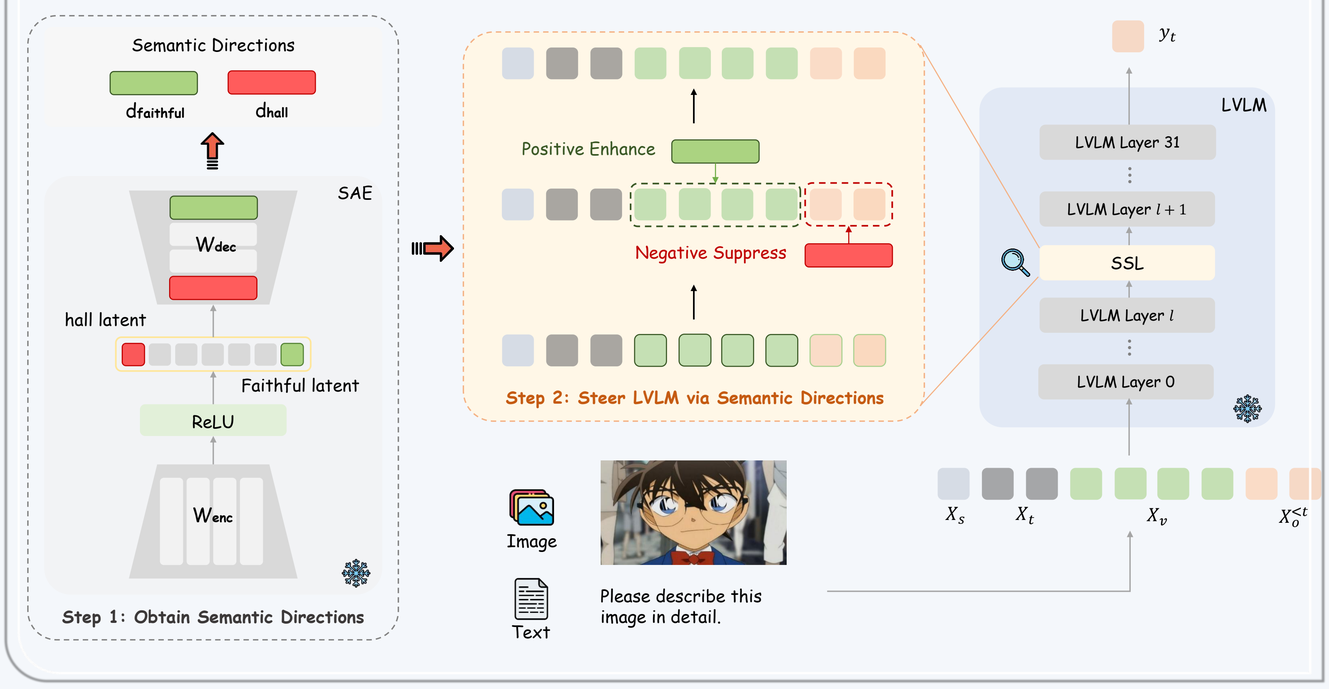
IDENTIFY SEMANTIC DIRECTIONS



Input hidden states into the SAE. The direction with the **largest activation difference** between hallucinatory and faithful tokens is hallucinatory direction, while the opposite direction is defined as the faithful direction.



METHODS -- SSL



SCAN FOR MORE

PAPER

CODE

WECHAT

EXPERIMENTS -- CHAIR & POPE

	LLaVA-NeXT-8b			LLaVA-1.5-7b			InstructBLIP-7b		
	$CHAIR_S \downarrow$	$CHAIR_I \downarrow$	*Avg.Len*	$CHAIR_S \downarrow$	$CHAIR_I \downarrow$	*Avg.Len*	$CHAIR_S \downarrow$	$CHAIR_I \downarrow$	*Avg.Len*
Greedy	$29.60_{\pm 0.89}$	$8.03_{\pm 0.41}$	165.61	$49.44_{\pm 1.57}$	$14.19_{\pm 0.76}$	82.97	$\underline{45.44}_{\pm 2.43}$	$\underline{13.07}_{\pm 0.71}$	92.11
Beam	$\underline{27.20}_{\pm 1.19}$	$\underline{7.20}_{\pm 0.39}$	174.17	$53.60_{\pm 2.39}$	$15.47_{\pm 0.45}$	87.38	$48.68_{\pm 1.65}$	$13.59_{\pm 0.43}$	95.92
DoLa	$29.04_{\pm 1.08}$	$7.86_{\pm 0.24}$	166.14	$50.64_{\pm 2.33}$	$14.51_{\pm 0.88}$	82.32	$46.12_{\pm 1.85}$	$13.09_{\pm 0.90}$	91.80
VCD	$31.36_{\pm 1.99}$	$8.40_{\pm 0.79}$	165.43	$51.68_{\pm 1.85}$	$15.29_{\pm 0.83}$	83.03	$50.84_{\pm 2.41}$	$14.51_{\pm 0.97}$	91.44
OPERA	-	-	-	$\underline{44.04}_{\pm 0.94}$	$\underline{13.23}_{\pm 0.46}$	75.79	$45.88_{\pm 2.31}$	$13.15_{\pm 0.87}$	93.51
CODE	$30.76_{\pm 0.92}$	$8.09_{\pm 0.42}$	158.07	$47.72_{\pm 0.79}$	$14.13_{\pm 0.56}$	78.43	$50.88_{\pm 2.05}$	$14.21_{\pm 0.92}$	89.62
SSL	$\mathbf{26.36}_{\pm 1.94}$	$\mathbf{6.32}_{\pm 0.63}$	163.86	$\mathbf{40.88}_{\pm 2.11}$	$\mathbf{12.30}_{\pm 1.18}$	82.37	$\mathbf{44.04}_{\pm 3.91}$	$\mathbf{12.64}_{\pm 1.38}$	100.26

	LLaVA-NeXT-8b F1 score ↑	LLaVA-1.5-7b F1 score ↑	InstructBLIP-7b F1 score ↑
Greedy	89.10	84.99	$\underline{85.37}$
Beam	89.30	85.31	84.41
DoLa	$\underline{89.49}$	85.08	85.22
VCD	88.91	84.42	84.68
OPERA	-	$\underline{85.46}$	84.42
CODE	88.93	84.64	84.81
SSL	**89.66**	**85.47**	**85.56**